

Matheus Leca

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Calgary, Alberta

Education

- **M.Sc., Electrical and Computer Engineering, University of Calgary, CAN** Graduated 2026
- **B.Sc., Computer Science, Federal University of Pernambuco, BRA** Graduated 2024

Work History

Software Engineer, University of Calgary, CAN

Apr 2025 – Aug 2026

- Developed a cross-platform **Unity 6 / C# safety-training simulator** for **web browsers, mobile, desktop, and Meta Quest VR**, engaging stakeholders to define and refine **functional and non-functional requirements** and integrating **HTML, CSS, and JavaScript** for web functionality to support immersive hazard identification, scoring, navigation, and 3D/360° workplace-safety training.
- Overcame a 100 MB WebGL hosting limit by engineering a custom asset pipeline with **automated asset splitting, parallel downloads, and client-side reassembly**, enabling deployment of the full simulator.
- Enabled online level management, sharing, leaderboards, and submission handling by engineering an **ASP.NET Core / .NET** backend with **Clean Architecture, EF Core, PostgreSQL**.
- Enabled reliable cloud-based simulator delivery through **DevOps practices**, using **Git** for version control, **GitHub** for collaboration, and **GitHub Actions** with **Continuous Integration/Continuous Delivery (CI/CD)** to **automate deployments, database migrations, and unit, integration, and end-to-end testing** for **Docker** infrastructure on **DigitalOcean** in **Linux/Unix** environments, using **Spaces + CDN** to serve 360° media.

Software Engineer, LogAp I.T. Solutions, BRA

Feb 2023 – Aug 2024

- Delivered enterprise **web and mobile applications** for government partners using **Java, Spring, Spring Boot, Maven, Gradle, C#, .NET, Angular, Ionic, HTML, CSS, JavaScript/TypeScript, and MSSQL**, helping digitize public services, reduce operational costs, and improve efficiency.
- Reduced **operational costs** by 18% at SGNF and 27% at BDGA by streamlining application workflows, collaborating with stakeholders to refine **functional and non-functional requirements**, and implementing **Spring microservices** and client-requested features, helping BDGA earn first place in “Regulatory Impacts” at EGAESE.
- Contributed to a 32% increase in Q1 revenue for Águas Brasil by digitizing business processes and integrating third-party services through **REST APIs**, using **RabbitMQ** for asynchronous messaging and **Kafka** for event-driven data exchange.
- Reduced API latency by 80% and processing turnaround time by 60% across ALIEN and HidroObserva by optimizing API processing and business logic using **Java and Hibernate**, applying **SOLID principles** to improve maintainability and extensibility, and deploying solutions in **Linux/Unix** environments. Ensured reliable delivery through **Git and GitHub**, with **GitHub Actions** automating **CI/CD and testing**, while following **Agile/Scrum** practices and conducting **code reviews** throughout the **Software Development Life Cycle (SDLC)**.

Software Engineer, Reciprev, BRA

Oct 2021 – Feb 2023

- Reduced social-security accounting software costs by 36% within the first year by collaborating with stakeholders to refine **functional and non-functional requirements** and digitizing and automating accounting processes through features built with **React, Vue, jQuery, Node.js, Express, HTML, CSS, JavaScript, and MongoDB**.
- Enabled reliable cloud deployment of the accounting application by hosting services on **AWS EC2** and managing application data with **Amazon RDS**, supporting scalable application hosting and persistent database management.
- Improved software reliability by applying **Test-Driven Development, unit and end-to-end testing**, and **Agile/Scrum** practices throughout the **SDLC**. Used **Git** for version control and **GitLab** for collaboration and **code reviews** to support reliable feature delivery.

Teaching Assistant, University of Calgary, CAN

Sep 2024 – Apr 2026

- Improved students' understanding of **software architecture and design patterns** through lectures and practical instruction on **microservices and modular monoliths** using **Java, Spring, Spring Boot, MySQL, MongoDB, HTML, CSS, and JavaScript**.
- Supported undergraduate and graduate instruction across 6 courses, helping students build skills in **programming, databases, software design, architecture, and web development**.
- Strengthened students' **object-oriented programming and data structures** skills through **Python** by providing individualized **guidance and feedback** on labs, assignments, and projects.

Teaching Assistant, Federal University of Pernambuco, BRA

Jul 2022 – Mar 2024

- Expanded access to AI-assisted learning for undergraduate students by developing **Turing Tutor**, integrating the **OpenAI Assistants API** into Discord using **JavaScript/TypeScript and Node.js**.
- Strengthened students' understanding of **Theoretical Foundations of Computer Science** through individualized **explanations, feedback, and academic guidance**.
- Improved learning and assessment by delivering lectures, creating and grading coursework, and holding weekly office hours to **explain technical concepts in plain language and provide individualized support**.

Projects

ThriveTree, Personal iOS Project

2026

- Developed a **Swift / SwiftUI** iOS app that gamifies habit building through a virtual tree, using animations, reminders, health states, and progress tracking to encourage consistent daily habits.

VeriFin, Personal Web Application

2026

- Developed a full-stack personal finance platform using **Java, Spring Boot, React, TypeScript, and PostgreSQL** to help users manage budgets, track expenses, set financial goals, and monitor investments.

Publications

- *FairPath: A Human-in-the-Loop Framework for Adversarial Fairness Testing in Machine Learning Pipelines*. Master's Thesis, 2026.
- *A Systematic Review of Bias Detection and Dataset Improvement Methods for Fair AI Systems*. FSE (HumanAISE '26), 2026.
- *Empathy in Software Engineering Education: Evidence, Practices, and Opportunities*. ICSE (CHASE '26), 2026.
- *Software Fairness Testing in Practice*. ICSME '25, 2025.
- *Curious, Critical Thinker, Empathetic, and Ethically Responsible: Essential Soft Skills for Data Scientists in Software Engineering*. ICSE (SEIS '25), 2025.
- *Towards User-Focused Cross-Domain Testing: Disentangling Accessibility, Usability, and Fairness*. SANER (SQA4AI '25), 2025.
- *Responsible AI in the Software Industry: A Practitioner-Centered Perspective*. ICSE (RAIE '25), 2025.
- *Applications and Implications of Large Language Models in Qualitative Analysis: A New Frontier for Empirical Software Engineering*. ICSE (WSESE '25), 2025.
- *Development and Evaluation of an Artificial Intelligence-Based Chatbot as a Learning Support Tool*. Bachelor's Thesis, 2023.